

Instructions for Organizers

Introduction

Overview

The DataStorm problem statement is great for shorter events with participants that are new to MATLAB. It provides them with a weather dataset and challenges them with 10 data extraction and analysis problems. We suggest you familiarize yourself with the problems and potential solutions ahead of time, as participants may have questions.

Audience and Event Timing

We recommend the following time allotment for this event:

- 1-2 hours for an audience that may have experience with coding or MATLAB
- 2-3 hours for participants that are brand new to coding and MATLAB.

The Dataset

This problem statement uses a dataset consisting of climate data from 2000-2010. The dataset has 6 parameters:

1. AirTemperature – degree celcius
2. RainfallFlux - mm/day
3. SnowfallFlux - mm/day
4. SpecificHumidity - g/kg
5. SurfAirPressure - hPa
6. WindSpeed - m/s

Each parameter is stored as an 11x365 matrix, with the rows representing the years, the columns representing the days, and the value at each (row,column) pair representing the measured data at that day on that year.

Name	Value	Size	Class
 AirTemperature	<i>11×365 double</i>	11×365	double
 RainfallFlux	<i>11×365 double</i>	11×365	double
 SnowfallFlux	<i>11×365 double</i>	11×365	double
 SpecificHumidity	<i>11×365 double</i>	11×365	double
 SurfAirPressure	<i>11×365 double</i>	11×365	double
 WindSpeed	<i>11×365 double</i>	11×365	double

Interested in using this problem statement?

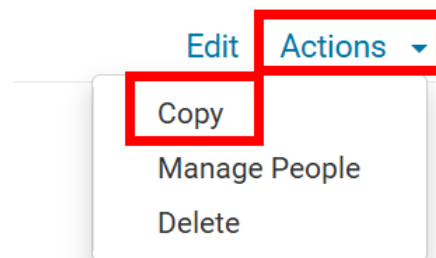
Reach out to hackathon@mathworks.com for access to the template materials in MATLAB Grader. You will need to meet the requirements for using MATLAB Grader:

- <https://www.mathworks.com/help/matlabgrader/ug/prerequisites-for-instructors.html>

Setup

Once you have access:

1. Go to <https://grader.mathworks.com/courses/167296>
2. Click on 'Actions' > 'Copy' on the right-hand corner

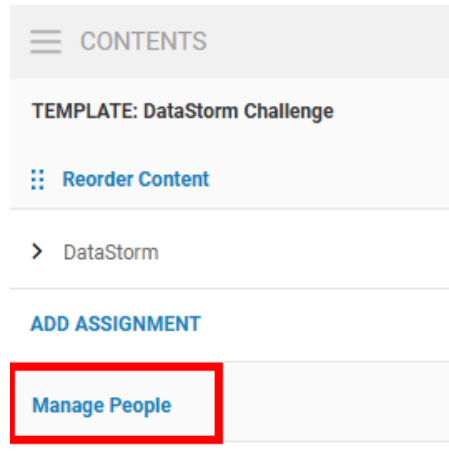


3. Click 'Copy' on the pop-up to create your own copy of this course, which should be called 'Copy of TEMPLATE: DataStorm Challenge' by default. This will open a new tab where you should do the following steps.
4. Name your event
5. Set the time zone
6. Set the start and end dates
7. Click 'Save'

Adding participants

Right before the event starts, you will need to add all participants to the MATLAB Grader course.

1. Go to grader.mathworks.com and click on the course you created earlier
2. Click on 'Manage People' on the left menu



- a.
3. In the page that opens, select 'Learner' in the drop down to enroll learners, then copy and paste all participants' emails. You can do these all at once.

Manage People

A screenshot of the 'Manage People' form. At the top right, there is an 'Enroll' dropdown menu. Below it, on the left, is another 'Enroll' dropdown menu with 'Learner' selected; this entire dropdown is highlighted with a red rectangular border. To the right of this dropdown is a blue circular help icon with a white question mark. Below the dropdowns is a large text input area containing the placeholder text 'participant1@email.com, participant2@email.com, participant3@email.com'. At the bottom right of the form are two buttons: 'Cancel' and 'Enroll'.

If you are using Devpost, follow [these instructions](#) for getting a list of participant emails. During the welcome presentation, you should tell participants to come up to you if they have not been added.

Evaluating

To export the results, click on the tab that says 'DataStorm':

MATLAB Grader

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ADD ASSIGNMENT

Manage People

TEMPLATE: DataStorm Challenge

Duration (EST): 13 Jun 2025 - 30 Jun 2025

Products:
Optimization Toolbox, Statistics and Machine Learning Toolbox, Symbolic

Course Description

Dive into the world of climate data analysis in MATLAB! Over the course of this

Then click on 'Actions' > 'Report' in the top-right corner

MATLAB Grader

Grace Woolson

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1. Annual Rainfall Trends

2. Pressure Extremes

3. Snowy Year

4. Climate Correlations

5. Wind Speed Patterns

6. Holiday!

7. Perfect Picnic!

Vite Elund

Women in Tech Trek: DataStorm Challenge

DataStorm

Visible: 28 Jul 2024 1:30 PM EDT Due: 29 Nov 2025 1:30 PM EST Submissions Per Assessment Item Unlimited

Assignment Description

Meet Memby, a young data scientist passionate about uncovering hidden patterns in nature. You have a mission to help Memby with specific tasks to uncover hidden patterns in Mathtown's weather data. Each challenge requires your data analysis skills to solve real-world problems.

The following is the climate data from the year 2000-2010. The dataset has 6 parameters:

- AirTemperature - degree celcius
- RainfallFlux - mm/day
- SnowfallFlux - mm/day
- SpecificHumidity - g/kg
- SurfAirPressure - hPa
- WindSpeed - m/s

The Climatedata.mat file is already loaded in the workspace. The mat file has the above 6 parameters in a 11x365 table each. Assume their are no leap years in the above data.

Edit

Actions

Copy

Report

Delete

On the popup that opens, choose to export the Best Solutions as of the due date as a CSV, then click 'Report'. This should download a .CSV file called 'DataStorm'.

Create an Assignment Report

Report includes, for each learner and assessment item:

- ☒ Best solution as of the due date (29 Nov 2025 1:30 PM EST)
- ☐ Best solution as of today
- ☐ All solutions

Output Format:

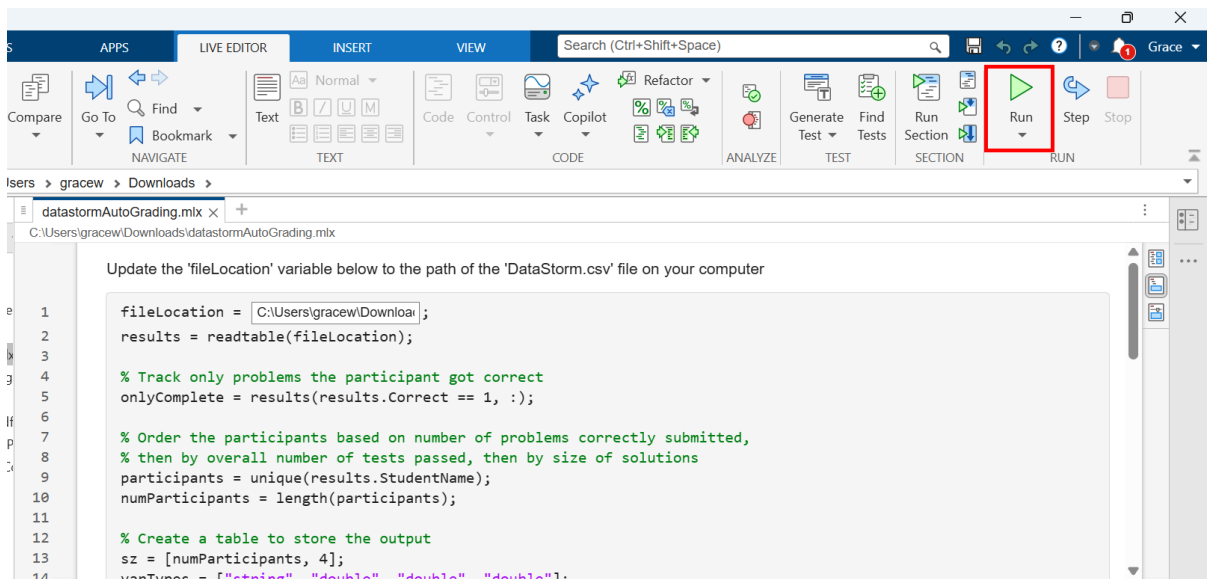
- ☒ CSV
- ☐ Excel
- ☐ MATLAB code files: Includes individual solution files, plus the reference solution and learner template. (Good for plagiarism detection software.)

Cancel

Report

Use this file with the MATLAB Live Script 'datastormAutoGrading.mlx' to determine the winners:

1. Open MATLAB
2. Open 'datastormAutoGrading.mlx'
3. Update the fileLocation to be the file path of the results (ex:
C:\Users\gracew\Downloads\DataStorm.csv)
4. Run the live script by clicking on the 'Run' Button



5. Scroll to the bottom and use the output to determine the winning participants/teams. This should be the first, second, and third rows of the table.